$e : \tau$	Condition	$\mathcal{D}[\![e : \tau]\!]$
$x : \tau$		x
c : **float**$[B; V]$	$B = \top$	c
	$B \neq \top$	$k_{\#(\lvert B\rvert+1)}$ such that $c \in \text{Intervals}(B)_k$
let $x = e_1 : \tau_1$ **in** $e_2 : \tau_2 : \tau_2$		**let** $x = \mathcal{D}[\![e_1 : \tau_1]\!]$ **in** $\mathcal{D}[\![e_2 : \tau_2]\!]$
(**if** e_1 : **bool then** $e_2 : \tau$ **else** $e_3 : \tau$) : τ		**if** $\mathcal{D}[\![e_1 : \textbf{bool}]\!]$ **then** $\mathcal{D}[\![e_2 : \tau]\!]$ **else** $\mathcal{D}[\![e_3 : \tau]\!]$
$(e_1$: **float**$[B; V_1] < e_2$: **float**$[B; V_2])$: **bool**	$B \neq \top$	$\mathcal{D}[\![e_1 : \textbf{float}[B; V_1]]\!] <_{\#(\lvert B\rvert+1)} \mathcal{D}[\![e_2 : \textbf{float}[B; V_2]]\!]$
	$B = \top$	$\mathcal{D}[\![e_1 : \textbf{float}[B; V_1]]\!] < \mathcal{D}[\![e_2 : \textbf{float}[B; V_2]]\!]$
$(e_1$: **float**$[B; V_1] \leq e_2$: **float**$[B; V_2])$: **bool**	$B \neq \top$	$\mathcal{D}[\![e_1 : \textbf{float}[B; V_1]]\!] \leq_{\#(\lvert B\rvert+1)} \mathcal{D}[\![e_2 : \textbf{float}[B; V_2]]\!]$
	$B = \top$	$\mathcal{D}[\![e_1 : \textbf{float}[B; V_1]]\!] \leq \mathcal{D}[\![e_2 : \textbf{float}[B; V_2]]\!]$
uniform$(e_1$: **float**$[B_1; V_1]$, e_2 : **float**$[B_2; V_2])$: **float**$[B; V]$	~~$e_1, e_2 \in \mathbb{R},\ e_1 > e_2$~~ $e_1, e_2 \in \mathbb{R},\ e_1 \geq e_2$	**diverge** : τ
	~~$B = \top$~~ $B = \top,\ \neg(e_1, e_2 \in \mathbb{R} \wedge e_1 \geq e_2)$	**uniform**$(\mathcal{D}[\![e_1 : \textbf{float}[B_1; V_1]]\!], \mathcal{D}[\![e_2 : \textbf{float}[B_2; V_2]]\!])$
	~~$e_1, e_2 \in \mathbb{R},$~~ $B = \{\sim_1 b_1, \ldots, \sim_n b_n\}$	~~e_1, e_2 **discrete**$(p_0, \ldots, p_n)$~~ ~~where $p_k = \text{CDF}(b_{k+1}) - \text{CDF}(b_k)$~~ **discrete**$(p_0, \ldots, p_n)$ where $p_k = \Pr[X \in \text{Intervals}(B)_k],\ X$
		let $v_1' = \mathcal{D}[\![e_1 : \textbf{float}[B_1; V_1]]\!]$ **in** **let** $v_2' = \mathcal{D}[\![e_2 : \textbf{float}[B_2; V_2]]\!]$ **in** **if** $v_1' ==_{\#(\lvert B_1\rvert+1)} \mathcal{D}[\![v_1 : \textbf{float}[B_1; V_1]]\!]$ && $v_2' ==_{\#(\lvert B_2\rvert+1)} \mathcal{D}[\![w_1 : \textbf{float}[B_2; V_2]]\!]$ **then** $\mathcal{D}[\![\textbf{uniform}(v_1, w_1) : \textbf{float}[B; V]]\!]$ ...
	~~$B \neq \top$,~~ ~~$B_1 \neq \top$,~~ ~~$B_2 \neq \top$,~~ $V_1 = \{v_1, \ldots, v_n\}$, $V_2 = \{w_1, \ldots, w_m\}$	~~$e_1 \notin \mathbb{R}$ or $e_2 \notin \mathbb{R}$)~~ **else if** $v_1' ==_{\#(\lvert B_1\rvert+1)} \mathcal{D}[\![v_i : \textbf{float}[B_1; V_1]]\!]$ $B \neq \top$, $B_1 \neq \top$, && $v_2' ==_{\#(\lvert B_2\rvert+1)} \mathcal{D}[\![w_j : \textbf{float}[B_2; V_2]]\!]$ **then** ~~$B_2 \neq \top$,~~ $\mathcal{D}[\![\textbf{uniform}(v_i, w_j) : \textbf{float}[B; V]]\!]$ $V_1 = \{v_1, \ldots, v_n\}$, $V_2 = \{w_1, \ldots, w_m\}$ ...
		else $\mathcal{D}[\![\textbf{uniform}(v_n, w_m) : \textbf{float}[B; V]]\!]$

Fig. 13. Discretization function. The displayed cases are pairwise disjoint. Uniform distributions require a strict parameter ordering $e_1 < e_2$. Equivalently, ~~where $\text{CDF}(b_0) = 0$~~ because such distributions are atomless, the interval masses in the constant-parameter rule can be computed from the clamped CDF, with $\text{CDF}(-\infty) = 0$ and ~~$\text{CDF}(b_{n+1}) = 1$~~$\text{CDF}(+\infty) = 1$. For the **uniform** distribution, $V = \top$ is implicit, and all remaining cut-set and value-set cases are ruled out by the constraints enforced during type inference. For all other language constructs, $\mathcal{D}[\![\cdot]\!]$ recurses structurally over them.

5 Discretization

This section presents local transformation rules for our type-directed discretization algorithm, with examples. Our goal is to compute, from a typed expression e : **bool** and its type annotations, a *discretized* expression $\mathcal{D}[\![e : \textbf{bool}]\!]$: **bool** such that e and $\mathcal{D}[\![e : \textbf{bool}]\!]$ are *semantically equivalent*, i.e., they induce the same distribution. We refer to this semantics-preserving transformation as *exact discretization*, to distinguish it from approaches [20, 28] that approximate continuous programs via externally chosen precision parameters. Exactness here is the equality proved by Theorem 6.1: the transformed program preserves the relevant probabilities of the source program, not merely an approximation with an error bound. Intuitively, when a program uses continuous quantities only through finitely many comparison outcomes, exact inference should reduce to a discrete problem, and exact discretization makes that reduction explicit.

Distribution	Density	Distribution	Density
Exponential	$\lambda e^{-\lambda x}$	Uniform	$\frac{1}{b-a}\mathbf{1}_{[a,b]}(x), a < b$
Laplace	$\frac{1}{2b}e^{-\frac{\lvert x-\mu\rvert}{b}}$	Gaussian	$\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
Cauchy	$\frac{1}{\pi\gamma\left[1+\left(\frac{x-x_0}{\gamma}\right)^2\right]}$	Beta	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$
Student-T	$c\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$	LogNormal	$\frac{1}{x\sigma\sqrt{2\pi}}e^{-\frac{(\ln x-\mu)^2}{2\sigma^2}}$
Chi-squared	$\frac{1}{2^{k/2}\Gamma(k/2)}x^{k/2-1}e^{-x/2}$	Gamma	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$
Logistic	$\frac{e^{-(x-\mu)/s}}{s(1+e^{-(x-\mu)/s})^2}$	Pareto	$\frac{\alpha x_m^\alpha}{x^{\alpha+1}}$
Rayleigh	$\frac{x}{\sigma^2}e^{-x^2/(2\sigma^2)}$	Weibull	$\frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1}e^{-(x/\lambda)^k}$
		Gumbel-1	$\frac{1}{\beta}e^{-(z+e^{-z})}, \quad z=\frac{x-\mu}{\beta}$
		Gumbel-2	$\frac{\alpha\beta x^{-\alpha-1}e^{-(x/\beta)^{-\alpha}}}{\beta}$
		Exponentiated Power	$k\lambda x^{k-1}e^{-(x/\lambda)^k}$
Discrete Distributions			
Poisson	$\frac{\lambda^k e^{-\lambda}}{k!}$	Geometric	$(1-p)^{k-1}p$
Binomial	$\binom{n}{k}p^k(1-p)^{n-k}$	Negative Binomial	$\binom{k+r-1}{k}(1-p)^r p^k$
Hypergeometric	$\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$	Bernoulli	$p^k(1-p)^{1-k}$
Discrete Uniform	$\frac{1}{n}$	Multinomial	$\frac{n!}{x_1!\cdots x_k!}p_1^{x_1}\cdots p_k^{x_k}$

- If $B = \top$: then $\mathcal{D}[\![c : \mathbf{float}[B;V]]\!] = c$ and $\mathcal{D}[\![\mathbf{float}[B;V]]\!] = \mathbf{float}$. Since $\mathrm{has}(V,c)$ still holds, c is well-typed at $\mathbf{float}$ by Float-Const.
- If $B \neq \top$: then $\mathcal{D}[\![c : \mathbf{float}[B;V]]\!] = k_{\#|B|+1}$ for the unique k such that $c \in \mathrm{Intervals}(B)_k$, and $\mathcal{D}[\![\mathbf{float}[B;V]]\!] = \mathbf{fin}(|B|+1)$. Since $0 \le k < |B|+1$, the finite constant $k_{\#|B|+1}$ is well-typed at $\mathbf{fin}(|B|+1)$ by FinConst.

Case $e = (\mathbf{let}\ x = e_1 : \tau_1\ \mathbf{in}\ e_2 : \tau_2)$. By the inductive hypothesis,

$$\mathcal{D}[\![\Gamma]\!] \vdash \mathcal{D}[\![e_1 : \tau_1]\!] : \mathcal{D}[\![\tau_1]\!] \quad \text{and} \quad \mathcal{D}[\![\Gamma, x : \tau_1]\!] \vdash \mathcal{D}[\![e_2 : \tau_2]\!] : \mathcal{D}[\![\tau_2]\!].$$

Note that $\mathcal{D}[\![\Gamma, x : \tau_1]\!] = \mathcal{D}[\![\Gamma]\!], x : \mathcal{D}[\![\tau_1]\!]$. Applying Let gives

$$\mathcal{D}[\![\Gamma]\!] \vdash \mathbf{let}\ x = \mathcal{D}[\![e_1 : \tau_1]\!]\ \mathbf{in}\ \mathcal{D}[\![e_2 : \tau_2]\!] : \mathcal{D}[\![\tau_2]\!].$$

Case $e = (\mathbf{if}\ e_1 : \mathbf{bool}\ \mathbf{then}\ e_2 : \tau\ \mathbf{else}\ e_3 : \tau)$. By the inductive hypothesis, $\mathcal{D}[\![\Gamma]\!] \vdash \mathcal{D}[\![e_i : \tau_i]\!] : \mathcal{D}[\![\tau_i]\!]$ for $i = 1,2,3$, where $\tau_1 = \mathbf{bool}$ and $\tau_2 = \tau_3 = \tau$. Since $\mathcal{D}[\![\mathbf{bool}]\!] = \mathbf{bool}$, applying If gives

$$\mathcal{D}[\![\Gamma]\!] \vdash \mathbf{if}\ \mathcal{D}[\![e_1 : \mathbf{bool}]\!]\ \mathbf{then}\ \mathcal{D}[\![e_2 : \tau]\!]\ \mathbf{else}\ \mathcal{D}[\![e_3 : \tau]\!] : \mathcal{D}[\![\tau]\!].$$

Case $e = (e_1 : \mathbf{float}[B;V_1] < e_2 : \mathbf{float}[B;V_2])$. The Less rule was applied, so both subexpressions share the same cut set B and $\mathrm{answers}_<(B,V_1,V_2)$ holds. By the inductive hypothesis, $\mathcal{D}[\![\Gamma]\!] \vdash \mathcal{D}[\![e_i : \mathbf{float}[B;V_i]]\!] : \mathcal{D}[\![\mathbf{float}[B;V_i]]\!]$ for $i = 1,2$.

- If $B = \top$: then $\mathcal{D}[\![\mathbf{float}[B;V_i]]\!] = \mathbf{float}$ for both i, and the standard float comparison gives type $\mathbf{bool} = \mathcal{D}[\![\mathbf{bool}]\!]$.
- If $B \neq \top$: then both subexpressions are discretized to type $\mathbf{fin}(|B|+1)$, and the finite comparison $<_{\#(|B|+1)}$ is well-typed at $\mathbf{bool} = \mathcal{D}[\![\mathbf{bool}]\!]$ by FinLess.

$\mathcal{D}[\![\Gamma]\!] \vdash \mathcal{D}[\![e_i : \mathbf{float}[B_i; V_i]]\!] : \mathcal{D}[\![\mathbf{float}[B_i; V_i]]\!]$ for $i = 1, 2$. We consider each sub-case of the discretization function:

- $e_1, e_2 \in \mathbb{R}$ and ~~$e_1 > e_2$~~$e_1 \geq e_2$: then $\mathcal{D}[\![\mathbf{uniform}(\ldots) : \mathbf{float}[B; \top]]\!] = \mathbf{diverge}$, which is well-typed at any type by Diverge.
- $B = \top$ and the literal-invalid condition above does not hold: then $\mathcal{D}[\![\mathbf{float}[B; \top]]\!] = \mathbf{float}$, and

$$\mathcal{D}[\![\mathbf{uniform}(e_1, e_2) : \mathbf{float}[B; \top]]\!] = \mathbf{uniform}(\mathcal{D}[\![e_1 : \mathbf{float}[B_1; V_1]]\!], \mathcal{D}[\![e_2 : \mathbf{float}[B_2; V_2]]\!]).$$

  Since recovers(B_1, V_1) and recovers(B_2, V_2) are preserved under discretization, the result is well-typed at $\mathbf{float}$ by Continuous-2.
- $B \neq \top$~~and~~, $e_1, e_2 \in \mathbb{R}$, and $e_1 < e_2$: then $\mathcal{D}[\![\mathbf{float}[B; \top]]\!] = \mathbf{fin}(|B| + 1)$, and

$$\mathcal{D}[\![\mathbf{uniform}(e_1, e_2) : \mathbf{float}[B; \top]]\!] = \mathbf{discrete}(p_0, \ldots, p_{|B|})$$

  where ~~$p_k = \mathrm{CDF}(b_{k+1}) - \mathrm{CDF}(b_k)$. Each~~ $p_k = \Pr[X \in \mathrm{Intervals}(B)_k]$ for $X \sim \mathrm{Uniform}(e_1, e_2)$. Hence the distribution is atomless, each $p_k \in [0, 1]$, and $\sum_k p_k = 1$, so this is well-typed at $\mathbf{fin}(|B| + 1)$ by Discrete.
- $B \neq \top, B_1 \neq \top, B_2 \neq \top$, and ~~$e_1$ or e_2 is a variable~~at least one of e_1, e_2 is nonliteral: the algorithm generates conditionals over all pairs (v_i, w_j), each branch dispatching to a constant parameter in $\mathbf{uniform}$. By the previous sub-case, each valid branch $\mathcal{D}[\![\mathbf{uniform}(v_i, w_j) : \mathbf{float}[B; \top]]\!]$ is well-typed at $\mathbf{fin}(|B| + 1) = \mathcal{D}[\![\mathbf{float}[B; \top]]\!]$, while equality or reversed bounds use the malformed branch. The conditional structure is well-typed by If (as in the $\mathbf{if}$ case above), so the full expression is well-typed at $\mathcal{D}[\![\mathbf{float}[B; \top]]\!]$.

Case e is any other language construct (functions, pairs, lists, *etc.*). By definition of $\mathcal{D}[\![\cdot]\!]$, the discretization recurses structurally over e and τ. That is, $\mathcal{D}[\![e : \tau]\!]$ is formed by applying $\mathcal{D}[\![\cdot]\!]$ to each subexpression and subtype of $e : \tau$, and $\mathcal{D}[\![\tau]\!]$ is formed by applying $\mathcal{D}[\![\cdot]\!]$ to each component type of τ. The result then follows immediately by applying the inductive hypothesis to each subterm, analogous to the above cases.

In every case, the typing derivation is preserved under $\mathcal{D}[\![\cdot]\!]$. Therefore,

$$\mathcal{D}[\![\Gamma]\!] \vdash \mathcal{D}[\![e : \tau]\!] : \mathcal{D}[\![\tau]\!]. \qquad \square$$

F Soundness Proof

F.1 Measure Theory Preliminaries

We first give a brief overview of basic measure-theoretic terms and lemmas, and refer the reader to standard textbooks for further background [43, 44].

Topological space. A topological space is a pair $(X, \mathcal{T})$ where $\mathcal{T}$ is a collection of subsets of X, called open sets, such that $\emptyset, X \in \mathcal{T}$, and $\mathcal{T}$ is closed under arbitrary unions and finite intersections.

σ-algebra. A σ-algebra on a set X is a collection Σ of subsets of X that includes X itself and is closed under complement and under countable unions. By implication, a σ-algebra includes the empty set $\emptyset$ and is closed under countable intersections. Given a topological space $(X, \mathcal{T})$, the Borel σ-algebra $\mathcal{B}(X)$ is the smallest σ-algebra containing all open sets in $\mathcal{T}$.

Measurable space. A measurable space is a pair (X, Σ) consisting of a set X and a σ-algebra Σ on X. When Σ is the Borel σ-algebra on a topological space X, we call (X, Σ) a Borel measurable space.

Measure. Let (X, Σ) be a measurable space. A function $\mu : \Sigma \to [0, \infty]$ is called a measure if it satisfies the following properties: nonnegativity, $\mu(\emptyset) = 0$, and countable additivity. When Σ is a Borel σ-algebra, μ is called a Borel measure.

$\mathbf{e} : \tau$	constraints	$[\![e : \tau]\!]$
$\mathbf{diverge} : \tau$		$\lambda A.0$
$v : \tau$		$\delta_{v:\tau}$
$\mathbf{let}\ x = e_1 : \tau_1\ \mathbf{in}\ e_2 : \tau_2 : \tau_2$	$e_1 = v$ e_1 no val.	$\delta_{e_2[v/x]:\tau_2}$ $[\![e_1 : \tau_1]\!] \gg\!= \lambda g.\delta_{\mathbf{let}\ x=g\ \mathbf{in}\ e_2:\tau_2:\tau_2}$
$\mathbf{if}\ e_1 : \mathbf{bool}\ \mathbf{then}\ e_2 : \tau\ \mathbf{else}\ e_3 : \tau : \tau$	$e_1 = \text{true}$ $e_1 = \text{false}$ o.w.	$\delta_{e_2:\tau}$ $\delta_{e_3:\tau}$ $[\![e_1 : \mathbf{bool}]\!] \gg\!= \lambda g.\delta_{\mathbf{if}\ g\ \mathbf{then}\ e_2:\tau\ \mathbf{else}\ e_3:\tau:\tau}$
$\mathbf{observe}\ (e : \mathbf{bool}) : \mathbf{unit}$	$e = \text{true}$ $e = \text{false}$ o.w.	$\delta_{\mathbf{unit}}$ $\delta_{\bot}$ $[\![e : \mathbf{bool}]\!] \gg\!= \lambda g.\delta_{\mathbf{observe}\ (g:\mathbf{bool}):\mathbf{unit}}$
$e_1 : \tau_1 < e_2 : \tau_2 : \mathbf{bool}$ for $\tau_1 = \mathbf{float}[B; V_1], \tau_2 = \mathbf{float}[B; V_2]$ or $\tau_1 = \tau_2 = \mathbf{fin}(n)$	$e_1 = v_1, e_2 = v_2, v_1 < v_2$ $e_1 = v_1, e_2 = v_2, v_1 \geq v_2$ $e_1 = v_1, e_2 \neq v_2$ o.w.	$\delta_{\text{true}:\mathbf{bool}}$ $\delta_{\text{false}:\mathbf{bool}}$ $[\![e_2 : \tau_2]\!] \gg\!= \lambda g.\delta_{v_1:\tau_1<g:\mathbf{bool}}$ $[\![e_1 : \tau_1]\!] \gg\!= \lambda g.\delta_{g<e_2:\tau_2:\mathbf{bool}}$
$\mathbf{uniform}(e_1 : \tau_1, e_2 : \tau_2) : \mathbf{float}[B; V]$	~~$e_1 = v_1, e_2 = v_2, v_1 \leq v_2$~~ $e_1 = v_1, e_2 = v_2, v_1 < v_2$ ~~$e_1 = v_1, e_2 = v_2, v_1 > v_2$~~ $e_1 = v_1, e_2 = v_2, v_1 \geq v_2$ $e_1 = v_1, e_2 \neq v_2$ o.w.	$\text{Uniform}(v_1, v_2) \gg\!= \lambda v.\delta_{(v:\mathbf{float}[B;V])}$ $[\![\mathbf{diverge} : \tau]\!]$ $[\![e_2 : \tau_2]\!] \gg\!= \lambda g.\delta_{\mathbf{uniform}(v_1:\tau_1,g):\tau}$ $[\![e_1 : \tau_1]\!] \gg\!= \lambda g.\delta_{\mathbf{uniform}(g,e_2:\tau_2):\tau}$
$\mathbf{discrete}(v_0, \ldots, v_{n-1}) : \mathbf{fin}(n)$	$\sum_i v_i = 1$	$\text{Discrete}(v_0, \ldots, v_{n-1})$
$\mathtt{DC}_{B,B_1,B_2,V_1,V_2}(e_1 : \tau_1, e_2 : \tau_2) : \mathbf{fin}(\lvert B\rvert + 1)$ $\tau = \mathbf{fin}(\lvert B\rvert + 1)$, $\tau_1 = \mathbf{fin}(\lvert B_1\rvert + 1), \tau_2 = \mathbf{fin}(\lvert B_2\rvert + 1)$, $V_1 = \{v_1, \ldots, v_n\}, V_2 = \{w_1, \ldots, w_m\}$	$e_1 = \mathcal{D}[\![v_i : \mathbf{float}[B_1; V_1]]\!]$, $e_2 = \mathcal{D}[\![w_j : \mathbf{float}[B_2; V_2]]\!]$ $e_1 = v', e_2 \neq w'$ o.w.	$[\![\mathcal{D}[\![\mathbf{uniform}(v_i, w_j) : \mathbf{float}[B; \top]]\!] : \tau]\!]$ $[\![e_2 : \tau_2]\!] \gg\!= \lambda g.\delta_{\mathtt{DC}_{B,B_1,B_2,V_1,V_2}(v':\tau_1,g)}$ $[\![e_1 : \tau_1]\!] \gg\!= \lambda g.\delta_{\mathtt{DC}_{B,B_1,B_2,V_1,V_2}(g,e_2:\tau_2)}$

Table 1. Small-step operational semantics. In the last case, the existence of suitable v_i and w_j is guaranteed since recovers(B_i, V_i) and $e_i \in V_i$ for $i \in \{1, 2\}$. Here, every function of the form $f(g) = \lambda g.\delta_{\ldots}$ is extended to $f_\bot$ in the sense of Definition F.29.

for every pairwise disjoint family $(A_i)_i$, and total mass $f(x)(Y) = 0 \leq 1$.

(2) Measurable function. For any $A \in \mathcal{F}_Y$, the map $x \mapsto f(x)(A) = 0$ is the constant-zero function on X, hence measurable.

Thus both conditions hold, so f is a Markov kernel. □

Lemma F.33 (Composition of Markov kernels). *Let (X, Σ_X), (Y, Σ_Y), and (Z, Σ_Z) be measurable spaces. If*

$$P : X \to \mathcal{D}(Y) \quad \textit{and} \quad Q : Y \to \mathcal{D}(Z)$$

are Markov kernels, then their composition

$$(P \gg\!= Q)(A) \;:=\; \int_{x \in supp(P)} Q(x)(A)\, P(dx)$$

defines a Markov kernel

$$P \gg\!= Q : X \to \mathcal{D}(Z).$$

Proof. See [30, Lemma 1.38]. □

F.4 Small-Step Semantics

Throughout this section, we denote by $e : \tau$ a well-typed expression of type τ and implicitly assume that all sub-expressions are annotated with types, as well. Moreover, for the typed expression $e : \tau = \mathbf{uniform}(e_1 : \mathbf{float}[B_1; V_1], e_2 : \mathbf{float}[B_2; V_2]) : \mathbf{float}[B, V]$ with $B, B_1, B_2 \neq \top$ and

$e : \tau$	constraints	$[\![e : \tau, \mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]]\!]$
$\mathbf{diverge} : \tau$		$\lambda A.0$
$v : \tau$		$\delta_{(v:\tau, \mathcal{D}[\![v:\tau]\!]:\mathcal{D}[\![\tau]\!])}$
$\mathbf{let}\ x = e_1 : \tau_1\ \mathbf{in}\ e_2 : \tau_2 : \tau_2$	$e_1 = v$ e_1 no val.	$\delta_{(e_2[v/x]:\tau_2, \mathcal{D}[\![e_2:\tau_2]\!][\mathcal{D}[\![v:\tau_1]\!]/x]:\mathcal{D}[\![\tau_2]\!])}$ $[\![e_1 : \tau_1, \mathcal{D}[\![e_1 : \tau_1]\!] : \mathcal{D}[\![\tau_1]\!]]\!]$ $\gg\!= \lambda(g, g').\delta_{(\mathbf{let}\ x=g\ \mathbf{in}\ e_2:\tau_2:\tau_2, \mathbf{let}\ x=g'\ \mathbf{in}\ \mathcal{D}[\![e_2:\tau_2]\!]:\mathcal{D}[\![\tau_2]\!]:\mathcal{D}[\![\tau_2]\!])}$
$\mathbf{if}\ e_1 : \mathbf{bool}\ \mathbf{then}\ e_2 : \tau\ \mathbf{else}\ e_3 : \tau : \tau$	$e_1 = \text{true}$ $e_1 = \text{false}$ o.w.	$\delta_{(e_2:\tau, \mathcal{D}[\![e_2:\tau]\!]:\mathcal{D}[\![\tau]\!])}$ $\delta_{(e_3:\tau, \mathcal{D}[\![e_3:\tau]\!]:\mathcal{D}[\![\tau]\!])}$ $[\![e_1 : \mathbf{bool}, \mathcal{D}[\![e_1 : \mathbf{bool}]\!] : \mathcal{D}[\![\mathbf{bool}]\!]]\!]$ $\gg\!= \lambda(g, g').\delta_{(\mathbf{if}\ g\ \mathbf{then}\ e_2:\tau\ \mathbf{else}\ e_3:\tau:\tau,\ \mathbf{if}\ g'\ \mathbf{then}\ \mathcal{D}[\![e_2:\tau]\!]:\mathcal{D}[\![\tau]\!]\ \mathbf{else}\ \mathcal{D}[\![e_3:\tau]\!]:\mathcal{D}[\![\tau]\!]:\mathcal{D}[\![\tau]\!])}$
$\mathbf{observe}\ (e : \mathbf{bool}) : \mathbf{unit}$	$e = \text{true}$ $e = \text{false}$ o.w.	$\delta_{(\mathbf{unit},\mathbf{unit})}$ $\delta_{\perp}$ $[\![e : \mathbf{bool}, \mathcal{D}[\![e]\!] : \mathbf{bool}]\!]$ $\gg\!= \lambda(g, g').\delta_{(\mathbf{observe}\ (g:\mathbf{bool}):\mathbf{unit}, \mathbf{observe}\ (g':\mathbf{bool}):\mathbf{unit})}$
$e_1 : \tau_1 < e_2 : \tau_2 : \mathbf{bool}$ for $\tau_1 = \mathbf{float}[B; V_1], \tau_2 = \mathbf{float}[B; V_2]$ $B = \top$	$e_1 = v_1, e_2 = v_2, v_1 < v_2$ $e_1 = v_1, e_2 = v_2, v_1 \geq v_2$ $e_1 = v_1, e_2 \neq v_2$ o.w.	$\delta_{(\text{true}:\mathbf{bool}, \text{true}:\mathbf{bool})}$ $\delta_{(\text{false}:\mathbf{bool}, \text{false}:\mathbf{bool})}$ $[\![e_2 : \tau_2, \mathcal{D}[\![e_2 : \tau_2]\!] : \mathcal{D}[\![\tau_2]\!]]\!]$ $\gg\!= \lambda(g, g').\delta_{(v_1:\tau_1 <_g:\mathbf{bool}, \mathcal{D}[\![v_1:\tau_1]\!]:\mathcal{D}[\![\tau_1]\!] <_{g'}:\mathcal{D}[\![\mathbf{bool}]\!])}$ $[\![e_1 : \tau_1, \mathcal{D}[\![e_1 : \tau_1]\!] : \mathcal{D}[\![\tau_1]\!]]\!]$ $\gg\!= \lambda(g, g').\delta_{(g <_{e_2:\tau_2:\mathbf{bool}, g' <_{\mathcal{D}[\![e_2:\tau_2]\!]:\mathcal{D}[\![\tau_2]\!]:\mathcal{D}[\![\mathbf{bool}]\!])}$
$e_1 : \tau_1 < e_2 : \tau_2 : \mathbf{bool}$ for $\tau_1 = \mathbf{float}[B; V_1], \tau_2 = \mathbf{float}[B; V_2]$ $B \neq \top$	$e_1 = v_1, e_2 = v_2, v_1 < v_2$ $e_1 = v_1, e_2 = v_2, v_1 \geq v_2$ $e_1 = v_1, e_2 \neq v_2$ o.w.	$\delta_{(\text{true}:\mathbf{bool}, \text{true}:\mathbf{bool})}$ $\delta_{(\text{false}:\mathbf{bool}, \text{false}:\mathbf{bool})}$ $[\![e_2 : \tau_2, \mathcal{D}[\![e_2 : \tau_2]\!] : \mathcal{D}[\![\tau_2]\!]]\!]$ $\gg\!= \lambda(g, g').\delta_{(v_1:\tau_1 <_g:\mathbf{bool}, \mathcal{D}[\![v_1:\tau_1]\!]:\mathcal{D}[\![\tau_1]\!] <_{\#(\lvert B\rvert+1)} g':\mathcal{D}[\![\mathbf{bool}]\!])}$ $[\![e_1 : \tau_1, \mathcal{D}[\![e_1 : \tau_1]\!] : \mathcal{D}[\![\tau_1]\!]]\!]$ $\gg\!= \lambda(g, g').\delta_{(g <_{e_2:\tau_2:\mathbf{bool}, g' <_{\#(\lvert B\rvert+1)} \mathcal{D}[\![e_2:\tau_2]\!]:\mathcal{D}[\![\tau_2]\!]:\mathcal{D}[\![\mathbf{bool}]\!])}$
$\mathbf{uniform}(e_1 : \tau_1, e_2 : \tau_2) : \mathbf{float}[B; V]$ $B = \top$	~~$e_1 = v_1 \leq v_2 = e_2$~~ ~~$e_1 = v_1 < v_2 = e_2$~~ ~~$e_1 = v_1 > v_2 = e_2$~~ ~~$e_1 = v_1 \geq v_2 = e_2$~~ $e_1 = v_1, e_2$ no val. e_1 no val.	$\text{Uniform}(v_1, v_2) \gg\!= \lambda v.\delta_{(v:\mathbf{float}[B;V], v:\mathbf{float}[B;V])}$ $[\![\mathbf{diverge} : \tau, \mathcal{D}[\![\mathbf{uniform}(v_1 : \tau_1, v_2 : \tau_2)]\!] : \mathcal{D}[\![\tau]\!]]\!]$ $[\![e_2 : \tau_2, \mathcal{D}[\![e_2]\!] : \mathcal{D}[\![\tau_2]\!]]\!]$ $\gg\!= \lambda(g, g').\delta_{(\mathbf{uniform}(v_1:\tau_1, g:\tau_2):\tau, \mathbf{uniform}(\mathcal{D}[\![v_1]\!]:\mathcal{D}[\![\tau_1]\!], g':\mathcal{D}[\![\tau_2]\!]):\mathcal{D}[\![\tau]\!])}$ $[\![e_1 : \tau_1, \mathcal{D}[\![e_1]\!] : \mathcal{D}[\![\tau_1]\!]]\!]$ $\gg\!= \lambda(g, g').\delta_{(\mathbf{uniform}(g:\tau_1, e_2:\tau_2):\tau, \mathbf{uniform}(g':\mathcal{D}[\![\tau_1]\!], \mathcal{D}[\![e_2]\!]:\mathcal{D}[\![\tau_2]\!]):\mathcal{D}[\![\tau]\!])}$
$\mathbf{uniform}(e_1 : \tau_1, e_2 : \tau_2) : \mathbf{float}[B; V]$ $B \neq \top$ $\tau_1 = \mathbf{float}[B_1 \neq \top, \{v_1, \ldots, v_n\}]$ $\tau_2 = \mathbf{float}[B_2 \neq \top, \{w_1, \ldots, w_m\}]$	~~$e_1 = e_1 \leq e_2 = e_2$~~ ~~$e_1 = c_1 < c_2 = e_2$~~ values ~~$e_1 = v_1 > v_2 = e_2$~~ ~~$e_1 = v_1 \geq v_2 = e_2$~~ $e_1 = c_1, e_2$ no val. e_1 no val.	$\text{Uniform}(c_1, c_2) \gg\!= \lambda v.\delta_{(v:\mathbf{float}[B;V], v_i:\mathbf{fin}(\lvert B\rvert+1))}$ where v_i is unique number s.t. $v \in \text{Intervals}(B)_{v_i}$ $[\![\mathbf{diverge} : \tau, \mathcal{D}[\![\mathbf{uniform}(v_1 : \tau_1, v_2 : \tau_2)]\!] : \mathcal{D}[\![\tau]\!]]\!]$ $[\![e_2 : \tau_2, \mathcal{D}[\![e_2]\!] : \mathcal{D}[\![\tau_2]\!]]\!]$ $\gg\!= \lambda(g, g').\delta_{(\mathbf{uniform}(v_i:\tau_1, g:\tau_2):\tau, \mathsf{DC}_{B, B_1, B_2, V_1, V_2}(\mathcal{D}[\![v_1]\!]:\mathcal{D}[\![\tau_1]\!], g':\mathcal{D}[\![\tau_2]\!]):\mathcal{D}[\![\tau]\!])}$ $[\![e_1 : \tau_1, \mathcal{D}[\![e_1]\!] : \mathcal{D}[\![\tau_1]\!]]\!]$ $\gg\!= \lambda(g, g').\delta_{(\mathbf{uniform}(g:\tau_1, e_2:\tau_2):\tau, \mathsf{DC}_{B, B_1, B_2, V_1, V_2}(g':\mathcal{D}[\![\tau_1]\!], \mathcal{D}[\![e_2]\!]:\mathcal{D}[\![\tau_2]\!]):\mathcal{D}[\![\tau]\!])}$

Table 3. Small-step coupling semantics. Notice that $\mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]$ is uniquely determined by $e : \tau$ so that it suffices to indicate the latter in the left-hand column. Here, every function of the form $f(g, g') = \lambda(g, g').\delta_{\ldots}$ is extended to $f_\perp$ in the sense of Definition F.29.

is a Markov kernel.
Thus, $[\![\mathbf{diverge} : \tau]\!] : Expr_\tau \to \mathcal{D}(Expr_\tau + \perp)$ is a Markov kernel.
- All other cases are analogous.

□

F.5 Small-Step Coupling Semantics

Define $P_\tau = \{(e : \tau, \mathcal{D}[\![e : \tau]\!] : \mathcal{D}[\![\tau]\!]) \mid e : \tau\}$. Exploiting that $e : \tau$ is a value iff $\mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]$ is a value, we define:

Definition F.40. The small-step coupling $[\![e : \tau, \mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]]\!] \in \mathcal{D}(P_\tau + \perp)$ is defined recursively on the structure of $e : \tau$ by the rules in Tables 3 and 4.

Lemma F.41 (Monadic bind is a Markov kernel). *For every occurrence of a monadic bind $\mu \gg\!= (f, f')$ given in the small-step operational coupling semantics in Tables 3 and 4, $(f, f') : (P_{\tau_1} + \perp) \to \mathcal{D}(P_{\tau_2} + \perp)$ is a Markov kernel.*

We now define suitable projection functions on measures $\mu \in \mathcal{D}(P_\tau + \bot)$ by

$$\pi_1(\mu) = \mu \gg\!= \lambda(e_1, e_2).\delta_{e_1} \tag{1}$$
$$\pi_2(\mu) = \mu \gg\!= \lambda(e_1, e_2).\delta_{e_2}\ . \tag{2}$$

We have the following key property:

Lemma F.43 (Small-step coupling projection). *Let $e : \tau$. We have:*

(1) $\pi_1([\![e : \tau, \mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]]\!]) = [\![e : \tau]\!]$.
(2) $\pi_2([\![e : \tau, \mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]]\!]) = [\![\mathcal{D}[\![e]\!] : \mathcal{D}[\![\tau]\!]]\!]$.

Proof. By induction on the type derivation of $e : \tau$. The most interesting case is

$$e : \tau = \mathbf{uniform}(c_1 : \mathbf{float}[B_1; V_1], c_2 : \mathbf{float}[B_2; V_2]) : \mathbf{float}[B; V]\ ,$$

where

(1) ~~$e_1 \le e_2$~~ $c_1 < c_2$,
(2) $B = \{\sim_1 b_1, \ldots, \sim_n b_n\}$,
(3) $V = \top$,
(4) $V_1 = \{v_1, \ldots, v_{n_1}\}$ with recovers(B_1, V_1) and $c_1 \in V_1$ due to has(V_1, c_1) by I.H.
(5) $V_2 = \{w_1, \ldots, w_{n_2}\}$ with recovers(B_2, V_2) and $c_2 \in V_2$ due to has(V_2, c_2) by I.H.

First observe that we have

$$\begin{aligned}
&\mathcal{D}[\![e : \tau]\!] : \mathcal{D}[\![\tau]\!] \\
&= \mathtt{DC}_{B,B_1,B_2,V_1,V_2}(\mathcal{D}[\![c_1 : \mathbf{float}[B_1; V_1]]\!], \mathcal{D}[\![c_2 : \mathbf{float}[B_2; V_2]]\!]) : \mathbf{fin}(|B| + 1) \\
&= \mathtt{DC}_{B,B_1,B_2,V_1,V_2}(k_1 : \mathbf{fin}(|B_1| + 1), k_2 : \mathbf{fin}(|B_2| + 1)) : \mathbf{fin}(|B| + 1)\ ,
\end{aligned}$$

where k_i is the unique number such that $c_i \in \text{Intervals}(B_i)_{k_i}$. By Table 1 and Figure 13, this yields

$$[\![\mathcal{D}[\![e : \tau]\!]]\!] = [\![\mathbf{discrete}(p_0, \ldots, p_n)]\!] \text{ where } p_k = \text{CDFUniform}(b_{k+1} c_1, c_2) - \text{CDF}(b\text{Intervals}(B)_k) = \frac{b_{k+1} - c_1}{c_2 - c_1} - \frac{b_k - c_1}{c_2 - c_1}$$

Here $c_1 < c_2$, and the uniform CDF is the clamped function

$$F_{c_1,c_2}(x) = \begin{cases} 0 & x \le c_1, \\ \dfrac{x - c_1}{c_2 - c_1} & c_1 < x < c_2, \\ 1 & x \ge c_2. \end{cases}$$

Thus each p_k is the difference of F_{c_1,c_2} at consecutive cut locations (with endpoints $-\infty$ and $+\infty$). Strict and non-strict cuts assign a boundary point to different adjacent intervals, but this does not change these masses because $c_1 < c_2$ makes the uniform distribution atomless.

For Lemma F.43.2, consider the following:

$$
\begin{aligned}
& \pi_1(\llbracket e : \tau, \mathcal{D}\llbracket e \rrbracket : \mathcal{D}\llbracket \tau \rrbracket \rrbracket) \\
&= \llbracket e : \tau, \mathcal{D}\llbracket e \rrbracket : \mathcal{D}\llbracket \tau \rrbracket \rrbracket \gg\!= \lambda(e_1, e_2).\delta_{e_1} && \text{(definition)} \\
&= (\text{Uniform}(c_1, c_2) \gg\!= \lambda v.\delta_{(v:\mathbf{float}[B;V], v_i:\mathbf{fin}(|B|+1))}) \gg\!= \lambda(e_1, e_2).\delta_{e_1} \\
& && \text{(Table 3, } v_i \text{ is unique number s.t. } v \in \text{Intervals}(B)_{v_i}) \\
&= \text{Uniform}(c_1, c_2) \gg\!= \lambda v.(\delta_{(v:\mathbf{float}[B;V], v_i:\mathbf{fin}(|B|+1))} \gg\!= \lambda(e_1, e_2).\delta_{e_1}) \\
& && \text{(monad assoc., } v_i \text{ is unique number s.t. } v \in \text{Intervals}(B)_{v_i}) \\
&= \text{Uniform}(c_1, c_2) \gg\!= \lambda v.\delta_{v:\mathbf{float}[B;V]} \\
&= \text{Uniform}(c_1, c_2) \\
&= \llbracket e : \tau \rrbracket\,. && \text{(Table 1)}
\end{aligned}
$$

For Lemma F.43.2, consider the following:

$$
\begin{aligned}
& \pi_2(\llbracket e : \tau, \mathcal{D}\llbracket e \rrbracket : \mathcal{D}\llbracket \tau \rrbracket \rrbracket) \\
&= \llbracket e : \tau, \mathcal{D}\llbracket e \rrbracket : \mathcal{D}\llbracket \tau \rrbracket \rrbracket \gg\!= \lambda(e_1, e_2).\delta_{e_2} && \text{(definition)} \\
&= (\text{Uniform}(c_1, c_2) \gg\!= \lambda v.\delta_{(v:\mathbf{float}[B;V], v_i:\mathbf{fin}(|B|+1))}) \gg\!= \lambda(e_1, e_2).\delta_{e_2} \\
& && \text{(Table 3, } v_i \text{ is unique number s.t. } v \in \text{Intervals}(B)_{v_i}) \\
&= \text{Uniform}(c_1, c_2) \gg\!= \lambda v.(\delta_{(v:\mathbf{float}[B;V], v_i:\mathbf{fin}(|B|+1))} \gg\!= \lambda(e_1, e_2).\delta_{e_2}) \\
& && \text{(monad assoc., } v_i \text{ is unique number s.t. } v \in \text{Intervals}(B)_{v_i}) \\
&= \text{Uniform}(c_1, c_2) \gg\!= \lambda v.\delta_{v_i:\mathbf{fin}(|B|+1)} \\
& && \text{(monad return, } v_i \text{ is unique number s.t. } v \in \text{Intervals}(B)_{v_i}) \\
&= \llbracket \mathbf{discrete}(p_0, \ldots, p_n) \rrbracket \text{ where } p_k = \text{CDFUniform}(b_{k+1} c_1, c_2) - \text{CDF}(b\text{Intervals}(B)_k) = \frac{b_{k+1} - c_1}{c_2 - c_1} - \frac{b_k - c_1}{c_2 - c_1} \\
& && \text{(by construction)} \\
&= \llbracket \mathcal{D}\llbracket e \rrbracket : \mathcal{D}\llbracket \tau \rrbracket \rrbracket\,. && \text{(Table 1)}
\end{aligned}
$$

Now, the remaining cases are either (i) trivial base cases or (ii) immediately follow from the I.H. and Lemma F.44 further below. We demonstrate this for the case

$$
e : \tau = \mathbf{uniform}(e_1 : \mathbf{float}[B_1; V_1], e_2 : \mathbf{float}[B_2; V_2]) : \mathbf{float}[B; V]\,,
$$

where

(1) $e_1 = c_1$ is a value but e_2 is no value,
(2) $B = \{\sim_1 b_1, \ldots, \sim_n b_n\}$,
(3) $V = \top$,
(4) $V_1 = \{v_1, \ldots, v_{n_1}\}$ with recovers(B_1, V_1) and $c_1 \in V_1$ due to has(V_1, c_1) by I.H.
(5) $V_2 = \{w_1, \ldots, w_{n_2}\}$ with recovers(B_2, V_2) and $c_2 \in V_2$ due to has(V_2, c_2) by I.H.